PRODUCT USER MANUAL

IBI_ANALYSISFORECAST_BGC_005_004

Issue: 4.2

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RECORD TABLE

Issue	Date	§	Description of Change	Author	Validated By
1.0	18/01/2017	All	**[EIS April 2017: IBI-V3 version]** Creation of the PUM document. First release of the IBI biogeochemical analysis and forecast product. Operational launch coincident with update of the physical product (**IBI-V3** version).	M.G. Sotillo T. Dabrowski A. Amo E. Gutknecht	E. Álvarez Fanjul
2.0	18/01/2018	All	**[EIS March 2018: IBI-V4 version]** New system version. Changes on river inputs, permanent burial and atmospheric pCO2. Operational launch coincident with update of the physical product. Update of the whole historical catalogue.	A. Amo T. Dabrowski E. Gutknecht P. Bowyer M.G. Sotillo	E. Álvarez Fanjul A. Melet
2.1	17/04/2019	All	**[EIS July 2019: IBI-V5 version]** New variables distributed: pH, DIC and $spCO_2$ New dataset for static files.	A. Amo	M.G. Sotillo
2.2	03/04/2020		**[EIS July 2020: IBI-V5 version]** Forecast horizon extended to +10 days.	R. Aznar	M.G. Sotillo
3.0	10/09/2020		**[EIS December 2020: IBI-V6 version]** New system version. Revised river inputs. Permanent burial function of bottom friction. CI and OBC from new global system. Effect of wave-current coupling in forecast mode. Operational launch coincident with update of the physical product. Update of the whole historical catalogue. NetCDF-4 file format. Renaming of product/datasets.	A. Amo E. Gutknecht	M.G. Sotillo
4.0	08/08/2022	All	**[EIS November 2022: IBI-V7 version]** New system version. New variable distributed: zooplankton. Operational launch coincident with update of the physical product. Update of the whole historical catalogue. Delivery of IBI BIO best estimate: additional hindcast run (coupled to the best and definitive IBI PHY analysis).	A. Amo E. Gutknecht R. Aznar	M.G. Sotillo
4.1	16/06/2023	II.4	**[EIS November 2023: IBI-V7version]**	A. Amo E. Gutknecht	S. Cailleau M. G. Sotillo

			Update to the new NRT GLO PHY/BGC open boundary conditions (1/12°).	R. Aznar	
4.2	31/05/2024	All	**[EIS November 2024: IBI-V8 version]** New variable distributed: light attenuation. Updated product mask.	L. Castrillo R. Aznar	S. Cailleau M. G. Sotillo

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GLOSSARY AND ABBREVIATIONS

AVHRR	Advanced Very High-Resolution Radiometer – Optical instrument onboard NOAA satellites
CF	Climate Forecast (convention for NetCDF)
CMEMS	Copernicus Marine Environment Monitoring Service now referred as Copernicus Marine Service
ECMWF	European Centre for Medium Range Weather forecast
GLO	Global
IBI	Iberia Biscay Ireland
Meridional Velocity	West to East component of the horizontal velocity vector
MFC	Monitoring and Forecasting Centre
MY	Multi Year
NRT	Near Real Time
NetCDF	Network Common Data Form
PUM	Product User Manual
QUID	Quality Information Document
SLA	Sea Level Anomalies
SSH	Sea surface height
Zonal Velocity	South to North component of the horizontal velocity vector

DATA ACCESS

After registration, you will be able to download our data. To assist you, our HelpCenter is available, and more specifically its section about download.

Information on operational issues on products and services can be found on our User Notification Service. If you have any questions, please contact us.

I. INTRODUCTION

I.1 Summary

This document is the Product User Manual (PUM) for the IBI analysis and forecast product **IBI_ANALYSISFORECAST_BGC_005_004** of the Copernicus Marine Service (also referred to as the CMEMS). It provides aggregated hindcast updated weekly with 10-day forecast. Analysis is generated on a weekly basis, replacing the previous hindcast. An archive of analysis covering two years up to real-time is available on the server of the Copernicus Marine Service.

This product is defined on a standard grid at **1/36** degree (approx. 2-3 km), extending from 19°W to 5°E and 26°N to 56°N.

The product is organized in the following datasets:

- **cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1D-m**: 3D daily mean fields of chlorophyll, oxygen, iron, nitrate, ammonium, phosphate, silicate, net primary production, phytoplankton carbon, dissolved inorganic carbon, pH and zooplankton; and 2D daily mean fields of euphotic zone depth and surface partial pressure of carbon dioxide.
- **cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1D-m :** 3D daily mean fields of light attenuation (kd).
- **cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1M-m:** monthly averaged fields of the same variables delivered in the corresponding daily mean dataset.
- **cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1M-m :** monthly averaged fields of the same variables delivered in the corresponding daily mean dataset
- **cmems_mod_ibi_bgc_anfc_0.027deg-3D_static:** the bathymetry, masks and coordinates files used to generate this IBI product.

The forecast product is generated once a week:

- the time series (hindcast for the previous day and 10 days forecast) is updated at the end of the operational production process.
- old obsolete forecasts are deleted from the server.

Additionally, twice a week a biogeochemical run coupled into an IBI ocean physical solution with data assimilation is executed, providing best estimates that replace hindcasts previously delivered, and build the definitive historic time series.

The IBI-MFC Production Unit (run by Nologin with the support of CESGA, in terms of supercomputing resources) is responsible of the generation and delivery of the product.

The product is available on-line and disseminated through the Marine Data Store of the Copernicus Marine Service after automatic and human quality controls. Files downloaded are in **NetCDF-4** format and follow **CF-1.8** convention.

The analysis and forecast system are described and evaluated in the Quality Information Document (QUID – see link in References):

I.2 History of changes

System Version (Project/Service)	Operational launch	End of operations	Novelties
IBI-V3 **EIS Apr 2017** **(Copernicus-1 Phase-I)**	19/04/2017	22/03/2018	First release of the IBI MFC BIO operational System. Operational launch coincident with update of the physical product (**IBI-V3** version).
IBI-V4 **EIS Mar 2018** **(Copernicus-1 Phase-I)**	22/03/2018	09/07/2019	Model upgrade. Monthly atmospheric pCO_2 are used at the surface boundary condition. Changes on river inputs: additional inputs of NO3 and PO4 from EEA. No permanent burial in the sediments. Operational launch coincident with update of the physical product. Update of the historic best estimates on the catalogue.
IBI-V5 **EIS Jul 2019** **(Copernicus-1 Phase-II)**	09/07/2019	07/07/2020	Distribution of three variables related to carbon cycle (surface pCO2, pH, DIC). New dataset for static files.
IBI-V5 **EIS Jul 2020** **(Copernicus-1 Phase-II)**	07/07/2020	15/12/2020	Forecast horizon extended in time, passing from +7 days to +10 days.
IBI-V6 **EIS Dec 2020** **(Copernicus-1 Phase-II)**	15/12/2020	29/11/2022	Model upgrade. New formulation for permanent burial to the sediments: efficiency of permanent burial is function of the bottom shear stress relative to a critical threshold. Revised PO4 river inputs from EEA. Initial and open boundary conditions from the new Global solution at ¼°.

			Effect of wave-current coupling in forecast mode. NetCDF-4 file format. Renaming of product/datasets. Operational launch coincident with update of the physical product. Update of the historic best estimates on the catalogue.
IBI-V7 **EIS Nov 2022** **(Copernicus-2 Phase-I)**	29/11/2022	30/11/2023	Model upgrade Distribution of zooplankton biomass in carbon. Delivery of BIO best estimate: additional hindcast run (coupled to the best and definitive IBI PHY analysis). Operational launch coincident with update of the physical product. Update of the historic best estimates on the catalogue.
IBI-V7 **EIS Nov 2023** **(Copernicus-2 Phase-I)**	30/11/2023	26/11/2024	Update to the new NRT GLO PHY/BGC open boundary conditions (1/12°).
IBI-V8 **EIS Nov 2024** **(Copernicus-2 Phase-I)**	26/11/2024	…	Distribution of diffuse attenuation coeff of light in sea water, averaged over the 400-500 nm range (light attenuation). Updated product mask. Update product name.

Table 1. Historical evolution of the IBI MFC NRT biogeochemical operational analysis and forecast system along the Copernicus Marine Service. Time period in operation, as well as its main novelties with respect to the previous release, is provided for each system version.

II DESCRIPTION OF THE PRODUCT SPECIFICATION

II.1 General Information

Product Lines	**IBI_ANALYSISFORECAST_BGC _001_004**	
Geographical coverage	IBI longitude: [-19, 5] deg latitude: [26, 56] deg	
Variables	**cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1D-m** & **cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1M-m** Mass Concentration of Chlorophyll in Sea Water Mole Concentration of Dissolved Oxygen in Sea Water Mole Concentration of Iron in Sea Water Mole Concentration of Nitrate in Sea Water Mole Concentration of Ammonium in Sea Water Mole Concentration of Phosphate in Sea Water Mole Concentration of Silicate in Sea Water Net Primary Productivity of Carbon Euphotic Zone Depth Mole Concentration of Phytoplankton Expressed as Carbon in Sea Water Surface Partial Pressure of Carbon Dioxide in Sea Water Dissolved Inorganic Carbon Sea Water pH Reported on Total Scale Mole Concentration of Zooplankton Expressed as Carbon in Sea Water **cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1D-m** & **cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1M-m** Diffuse attenuation coeff of light in sea water, averaged over the 400-500 nm range **cmems_mod_ibi_bgc_anfc_0.027deg-3D_static** Bathymetry Mask Coordinates (e1t, e2t, e3t)	
Product Type	Analyse and Forecast	
Analysis / Update frequency	no	weekly (Mon & Wed)
	The IBI BIO model system has no data assimilation. So, no IBI analysis information is provided.	

	Nevertheless, the IBI BIO estimates are derived from a biogeochemical run coupled into an IBI ocean physical solution with data assimilation. An aggregation of this IBI BIO best estimates is delivered as historic time series.	
Forecast / Update frequency	yes	weekly (Thu)
Available time series	-2Y + 10-days forecast	
Temporal resolution	Daily (25hr averaged) 3D and 2D fields. Monthly 3D and 2D fields.	
Horizontal resolution	1/36º	
Number of vertical levels	50	
Target delivery time	1-day hindcast + 10-days forecast: Thursdays at 18:00 UTC. Monthly file for month-1 is available the 20th of every month.	
Format	NetCDF-4 CF1.8	
Delivery mechanism	Copernicus Marine Toolbox	

Table 2. General Information of the IBI_ANALYSISFORECAST_BGC_005_004 product.

II.2 Details of datasets

IBI_ANALYSISFORECAST_BGC_005_004
Datasets: **cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1D-m:** contains 3D and single-level daily averaged fields. **cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1M-m:** contains monthly averaged fields of the parameters distributed in the corresponding daily dataset. **cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1D-m :** contains daily mean fields of light attenuation (kd). **cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1M-m :** containing monthly averaged fields of the same variables delivered in the corresponding daily mean dataset. **cmems_mod_ibi_bgc_anfc_0.027deg-3D_static: contains coordinates, mask and bathymetry files.**

These datasets are composed of post-processed data into a regular 1/36º lat/lon grid that goes from 26°N to 56°N in latitude and 19°W to 5°E in longitude, called IBI service domain. Latitude and longitude step is 0.02778°, and the resulting horizontal grid extends to 1081 x 865 grid points. Information from all variables contained in these datasets is provided at the same grid points. The **daily** and **monthly** datasets provide daily and monthly averaged fields for the whole water column (except for 2D variables), and the whole IBI service domain. The static dataset contains the static files used for the generation of this IBI product. **Variables name in the NetCDF file and Unit: Long_name & Standard_name**
cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1D-m & cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1M-m
chl [mg.m-3] Mass Concentration of Chlorophyll in Sea Water mass_concentration_of_chlorophyll_a_in_sea_water
o2 [mmol.m-3] Mole Concentration of Dissolved Oxygen in Sea Water mole_concentration_of_dissolved_molecular_oxygen_in_sea_water
fe [mmol.m-3] Mole Concentration of Iron in Sea Water mole_concentration_of_dissolved_iron_in_sea_water
no3 [mmol.m-3] Mole Concentration of Nitrate in Sea Water mole_concentration_of_nitrate_in_sea_water
nh4 [mmol.m-3] Mole Concentration of Ammonium in Sea Water mole_concentration_of_ammonium_in_sea_water
po4 [mmol.m-3] Mole Concentration of Phosphate in Sea Water mole_concentration_of_phosphate_in_sea_water
si [mmol.m-3] Mole Concentration of Silicate in Sea Water mole_concentration_of_silicate_in_sea_water
nppv [mg.m-3.day-1] Net Primary Productivity of Carbon net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water

zeu [m] Euphotic Zone Depth euphotic_zone_depth
phyc [mmol.m-3] Mole Concentration of Phytoplankton Expressed as Carbon in Sea Water mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water
spco2 [Pa] Surface Partial Pressure of Carbon Dioxide in Sea Water surface_partial_pressure_of_carbon_dioxide_in_sea_water
dissic [mol.m-3] Dissolved Inorganic Carbon mole_concentration_of_dissolved_inorganic_carbon_in_sea_water
ph [] Sea Water pH Reported on Total Scale sea_water_ph_reported_on_total_scale
zooc [mmol.m-3] Mole Concentration of Zooplankton Expressed as Carbon in Sea Water mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water
cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1D-m & **cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1M-m**
kd [m-1] Diffuse attenuation coeff of light in sea water, averaged over the 400-500 nm range Volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water
cmems_mod_ibi_bgc_anfc_0.027deg-3D_static
IBI-MFC_005_004_mask_bathy.nc **deptho [m]** Bathymetry sea_floor_depth_below_geoid **mask** [] Land-sea mask: 1 = sea ; 0 = land sea_binary_mask
IBI-MFC_005_004_coordinates.nc **e1t [m]** Cell dimension along X axis

e2t [m] Cell dimension along Y axis **e3t [m]** Cell dimension along Z axis

Table 3. List of the datasets and variable names and unit for the IBI_ANALYSISFORECAST_BGC_005_004 product.

II.3 Additional Information on parameters

chl [mg.m-3]	total chlorophyll concentration in phytoplankton: sum of chlorophyll contents in nano- and micro-phytoplankton.
phyc [mmol.m-3]	total carbon content in phytoplankton: sum of carbon contents in nano- and micro-phytoplankton.
zooc [mmol.m-3]	Total carbon content in zooplankton : sum of carbon contents in micro- and meso-zooplankton.
kd[m-1]	Downwelling radiation is radiation from above. It does not mean "net downward". Radiative flux is the sum of shortwave and longwave radiative fluxes. When thought of as being incident on a surface, a radiative flux is sometimes called "irradiance". In addition, it is identical with the quantity measured by a cosine-collector light-meter and sometimes called "vector irradiance". In accordance with common usage in geophysical disciplines, "flux" implies per unit area, called "flux density" in physics. The volume scattering/absorption/attenuation coefficient is the fractional change of radiative flux per unit path length due to the stated process. Coefficients with canonical units of m2 s-1 i.e. multiplied by density have standard names with specific_ instead of volume_. The scattering/absorption/attenuation coefficient is assumed to be an integral over all wavelengths, unless a coordinate of radiation_wavelength is included to specify the wavelength. Attenuation is the sum of absorption and scattering. Attenuation is sometimes called "extinction". Also called "diffuse" attenuation, the attenuation of downwelling radiative flux refers to the decrease with decreasing height or increasing depth of the downwelling component of radiative flux, regardless of incident direction.

Table 4 : Details on the calculation of variables.

II.4 Production System Description

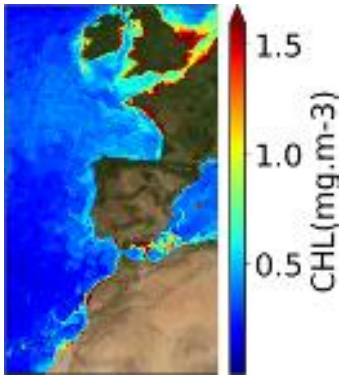
Domain	IBI (19°W-5°E; 26°N – 56°N)
Resolution and grid	1/36°; regular grid; 1081 x 865
Geographic coverage	This product is regional. It is defined on a regular grid at 1/36 degree (approx. 2-3 km) that goes from 26°N to 56°N in latitude and 19°W to 5°E in longitude. Latitude and longitude step is 0.02778°. The resulting horizontal grid extends to 1081x865 grid points. 
Model version	PISCES v2, part of the NEMO 3.6 modelling platform
Horizontal resolution	1/36°
Vertical coord.	50 levels
Atmospheric forcings	Hourly ECMWF (10-m wind, surface pressure, 2-m temperature, 2-m specific humidity, precipitation rates, short-wave and long-wave radiative fluxes).
Boundary conditions	Global ocean analysis and forecasting product at 1/12º (**GLOBAL_ANALYSISFORECAST_BGC_001_028**)
Initial conditions	Global ocean analysis and forecasting product at 1/12º (**GLOBAL_ANALYSISFORECAST_BGC_001_028)**
Bathymetry	GEBCO 08 (30 arc-second resolution).
Assimilation scheme	No (coupled into an IBI ocean physical solution with data assimilation).
Assimilated observations	No

Table 5. Summary of the system characteristics.

Short description

The IBI-MFC provides a high-resolution biogeochemical analysis and forecast product covering the European waters, and more specifically the Iberia–Biscay–Ireland (IBI) area. The last 2 years before now (historic best estimates) as well as daily averaged forecasts with a horizon of 10 days (updated on a weekly basis) are available on the catalogue.

To this aim, an online coupled physical-biogeochemical operational system is based on NEMO-PISCES at 1/36° and adapted to the IBI area, being Mercator-Ocean in charge of the model code development. PISCES is a model of intermediate complexity, with 24 prognostic variables. It simulates marine biological productivity of the lower trophic levels and describes the biogeochemical cycles of carbon and of the main nutrients (P, N, Si, Fe).

The product provides daily and monthly averages of the main biogeochemical variables: chlorophyll, oxygen, nitrate, phosphate, silicate, iron, ammonium, net primary production, euphotic zone depth, phytoplankton carbon, pH, dissolved inorganic carbon, surface partial pressure of carbon dioxide, zooplankton, and light attenuation.

Detailed description.

Biogeochemical model PISCES

The oceanic passive tracers of the ecosystem dynamics for lower trophic levels are simulated by the PISCES biogeochemical model (Pelagic Interactions Scheme for Carbon and Ecosystem Studies) in its v2 version (**Aumont et al., 2015**).

PISCES is a model of intermediate complexity, with 24 prognostic variables. It simulates marine biological productivity of the lower trophic levels and describes the biogeochemical cycles of carbon and of the main nutrients (P, N, Si, Fe). It considers five nutrients that limit phytoplankton growth (nitrate, ammonium, phosphate, silicate, and iron) and four living compartments: two phytoplankton size classes (nanophytoplankton and diatoms) and two zooplankton size classes (microzooplankton and mesozooplankton); the bacterial pool is not explicitly modelled. PISCES distinguishes three non-living detrital pools for organic carbon, particles of calcium carbonate and biogenic silicate. In addition, PISCES simulates the carbonate system and dissolved oxygen.

The standard “operational” version used here is described in **Figure 1** and **Aumont et al. (2015)**.

Biogeochemical parameters are based on the standard parameters of PISCES v2.

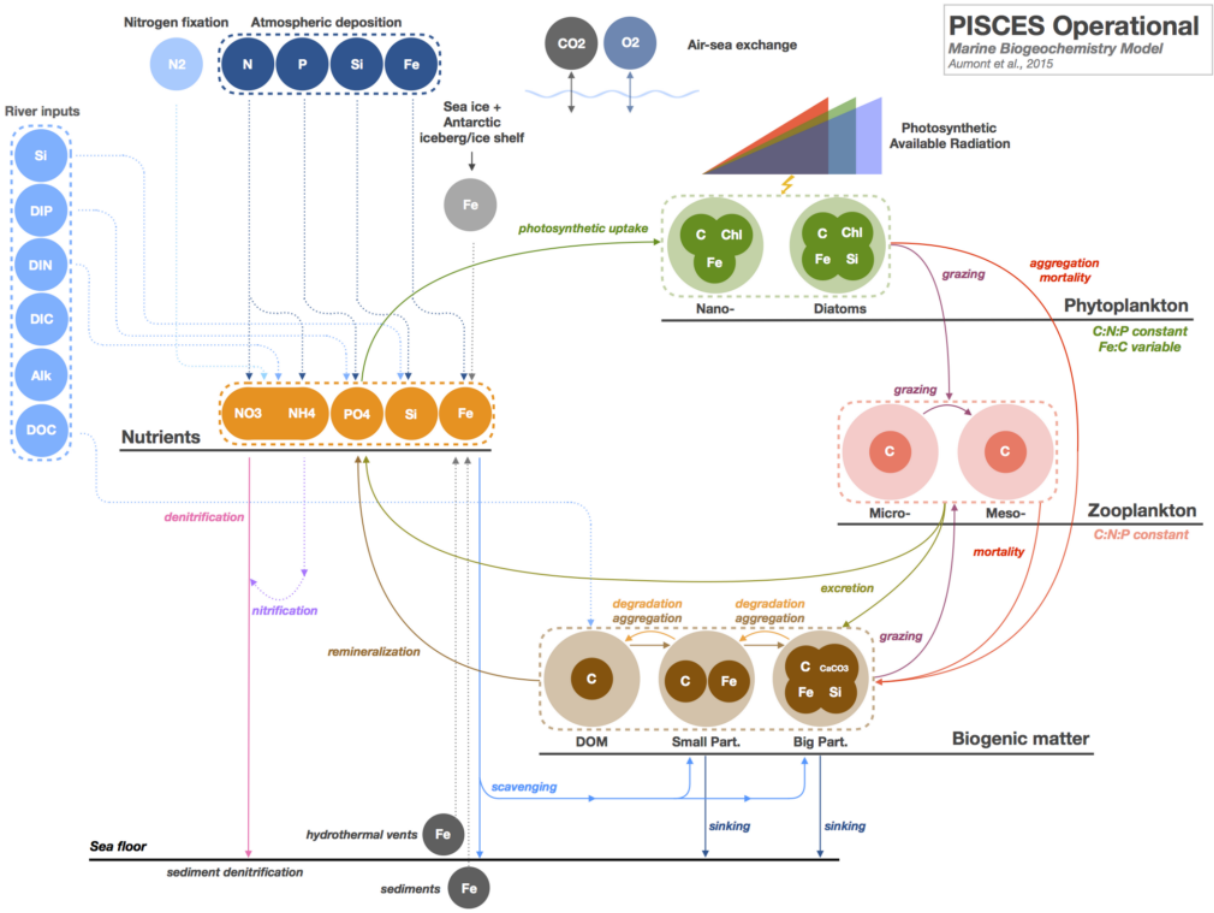


Figure 1. Schematic diagram of the standard "Operational" version of PISCES describing the nutrients, carbon and oxygen cycles and processes modelled. Source: https://www.pisces-community.org

Even if PISCES was initially designed for global ocean applications, the distinction of two phytoplankton size classes, along with the description of multiple nutrient co-limitations allows the model to represent ocean productivity and biogeochemical cycles across major biogeographic ocean provinces (**Longhurst, 1998**). PISCES has been successfully used in various biogeochemical studies at global, regional scales and up to process studies, at low and high spatial resolutions as well as for short-term and long-term analyses (e.g. **Bopp et al., 2005**; **Gehlen et al., 2006, 2007**; **Schneider et al., 2008**; **Steinacher et al., 2010**; **Tagliabue et al., 2010**, **Séférian et al., 2013**; **Gutknecht et al., 2016**).

PISCES is also used in operational oceanography (**Brasseur et al., 2009**) for the Indonesian seas operational system at 1/12° (the INDESO project; **Gutknecht et al., 2016**), and 4 products delivered to the Copernicus Marine Service: the global ocean analysis and forecast product at 1/4° (**GLOBAL_ANALYSIS_FORECAST_BIO_001_028**), the global ocean non-assimilative hindcast product at 1/4° (**GLOBAL_REANALYSIS_BIO_001_029**), the IBI non-assimilative hindcast product at 1/12° (**IBI_MULTIYEAR_BGC_005_003**) and the system covered by this document, the IBI analysis and forecast product at 1/36° (**IBI_ANALYSISFORECAST_BGC_005_004)**.

Physical model NEMO

The IBI ocean dynamics and thermodynamics are resolved by version 3.6 of the NEMO modelling platform (Nucleus for European Modelling of the Ocean; **Madec et al., 1998; Madec, 2008**) at 1/36 ◦ spatial resolution. Specific numerical choices include time-splitting and non-linear free surface to correctly simulate small-scale phenomena and high-frequency processes of primary importance such as tides and atmospheric pressure forcing.

The IBI36 physical component is described in **Maraldi et al. (2013), Sotillo et al. (2015)**, as well as in the Products User Manual Document **CMEMS-IBI-PUM-005-001** and the QUality Information Document **CMEMS-IBI-QUID-005-001**.

Coupling

PISCESv2 is part of the NEMO 3.6 modelling platform. PISCES is coupled to NEMO-OPA via the TOP component that manages the advection/diffusion equations of passive tracers but also the sources and sinks terms due to biogeochemistry. For this regional configuration, physics and biogeochemistry are running simultaneously ("on-line" coupling), with the same 1/36° horizontal resolution (~2km).

The ocean physics is coupled with the IBI wave component (**IBI_ANALYSIS_FORECAST_WAV_005_005** product) only in forecast mode. The pre-operational qualification simulation (see the QUality Information **Document CMEMS-IBI-QUID-005-004** for more details), as well as the 2 years historic best estimates available in the catalogue of the Copernicus Marine Service are therefore uncoupled.

Numerical choices

For reason of numerical cost (optimization of the computing time), the numerical scheme for biogeochemical processes is forward in time (Euler), while the physical component uses the leap-frog scheme. To respect the mass conservation, the coupling between biogeochemical and physical components is done every other time. The time step for the biogeochemical model is therefore twice that of the physical component, i.e., 300 s. The advection scheme for biogeochemistry is the same QUICKEST scheme (**Leonard, 1979**) with the limiter of **Zalesak (1979)** used for the physical part.

Initialisation and open boundary conditions

The biogeochemistry is initialized with the NEMO-PISCES global ocean analysis and forecasting system (**Perruche et al., 2016**) delivered in the Copernicus Marine Service (**GLOBAL_ANALYSIS_FORECAST_BIO_001_028** product) for the same date. Open boundary conditions come from the same global product on a weekly basis.

This pre-operational qualification simulation has served as initial conditions to start the operational analysis and forecast system, with the approach at open boundary conditions.

Atmospheric pCO2

The carbon cycle takes into account the anthropogenic effects, with real condition for atmospheric CO_2 partial pressure. Monthly mean carbon dioxide globally averaged over marine surface sites come from **Conway et al. (1994)** and **Masarie and Tans (1995)** available at https://www.esrl.noaa.gov/gmd/ccgg/trends/global.html

External supply of nutrients and carbon

Other boundary fluxes account for the external supply of nutrients and carbon from three different sources:

- The model includes an atmospheric dust deposition of Fe, Si, P and N at the surface ocean (**Aumont et al., 2015**).

- An iron source corresponding to sediment reductive mobilisation on continental margins is also considered (**Aumont et al., 2015**).

- River discharge of carbon comes from **Ludwig et al. (1996**). River discharge of nutrients comes from the Global NEWS 2 dataset (**Mayorga et al., 2010**) and the European Environment Agency (EEA; www.eea.europa.eu). Global NEWS 2 is an annual climatology at ½° spatial resolution which represents a realistic hydrology for the reference year 2000, considered as representative of the contemporary conditions (**Mayorga et al., 2010**). However, Global NEWS 2 seems to underestimate nutrient runoffs in the Western Europe (**Mayorga et al., 2010**). So additional inputs of nitrate and phosphate are injected into the IBI system to consider more recent values corresponding to the current decade. They come from rivers monitored and listed by the EEA based on annual averages.

Regional adjustment

The regional adaptation concerns the becoming of particles reaching the bottom boundary. Within the standard version of PISCES v2, the exchange between the ocean and the sediments assumes that a fraction of particulate material reaching the sea floor is permanently buried in the sediments while the remaining organic matter is dissolved or degraded and released into the water column. Concerning the IBI configuration, strong tidal currents prevent organic matter from settling on the bottom and being stored in the sediments over much of the North-West European continental shelf (**De Haas et al., 2002**). The IBI system then adjusts the efficiency of permanent burial based on the bottom shear stress relative to a critical threshold (0.1 N/m^2).

Data assimilation

The ocean physics is constrained through data assimilation of in situ and satellite physical data (temperature and salinity vertical profiles, sea surface height and sea surface temperature). The data assimilation method is described in **CMEMS-IBI-PUM-005-001.**

The biogeochemistry is not constrained through data assimilation but is nested to the global ocean analysis and forecasting system which is constrained through data assimilation of ocean colour. Please refer to **GLOBAL_ANALYSIS_FORECAST_BIO_001_028** product for more details. So IBI36 indirectly enjoy from data assimilation.

Catalogue generation.

The IBI BIO model system has no **data assimilation**. So, no IBI analysis information is provided. Nevertheless, the IBI BIO estimates come from a biogeochemical run coupled with the IBI ocean physical solution that is constrained through data assimilation of in situ and satellite physical data (temperature and salinity vertical profiles, sea surface height and sea surface temperature), and forced with ECMWF atmospheric analysis. An aggregation of these IBI BIO best estimates is delivered as historic time series of the **IBI_ANALYSISFORECAST_BGC_005_004.**

This BIO-PHY coupled run is executed twice a week. A first run is performed on Wednesdays (day D0), generating the D0-7 to D0-1 temporary solution. Two Mondays later (T0+12), this same period is re-run again, but this time nesting the IBI model into the best analysis from the GLOBAL system of the Copernicus Marine Service. The resulting product from this latest IBI run is what is kept as historical IBI best estimates.

Together with this concatenation of IBI best estimates (generated on a weekly basis), forecast information is delivered to Users from a D0+10 days run performed each Thursday. The conceptual scheme for the generation of the IBI BIO catalogue can be seen in **Figure 2**.

Finally, it is important to notice that only **the last 2 years** of long time series of historical best estimates **are available on the catalogue**, i.e., minus 2 years up to present time.

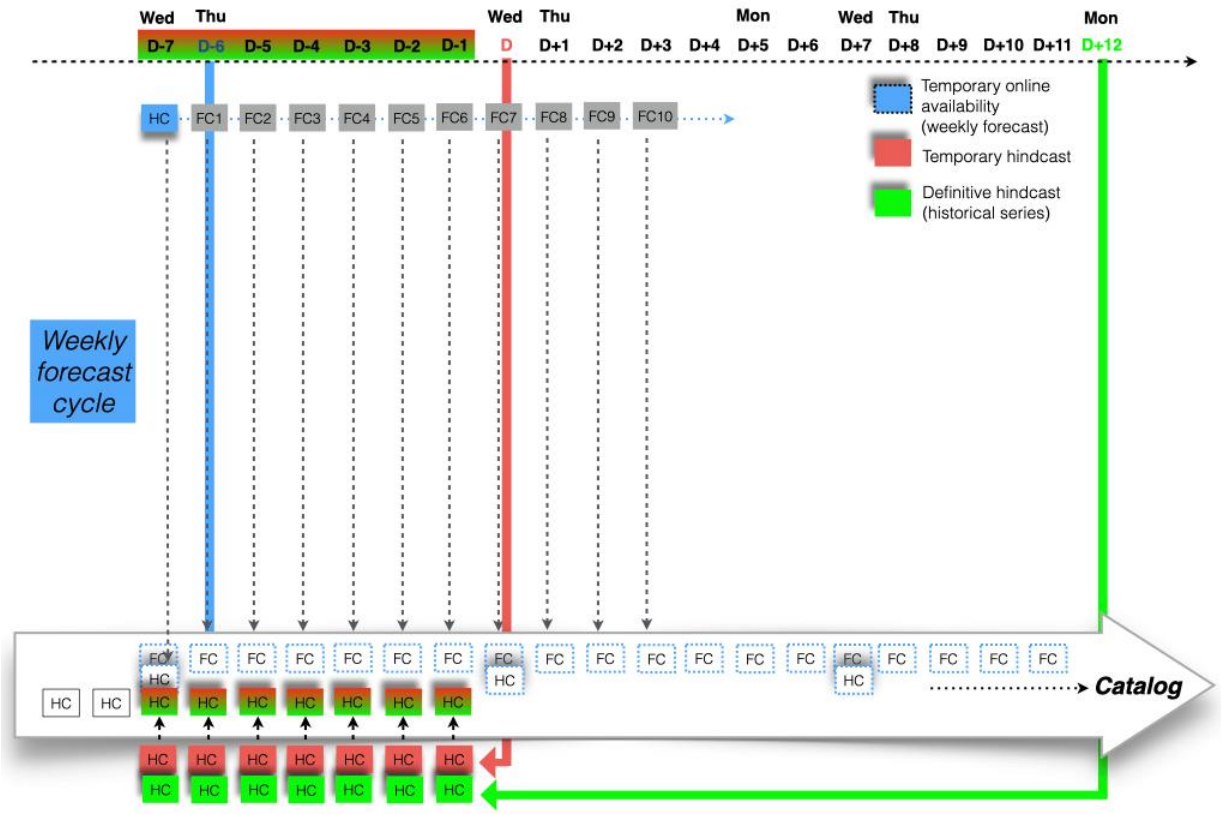


*Figure 2. Conceptual scheme for generation of the IBI BGC analysis and forecast product catalogue. A 10-day forecast run is executed on a weekly basis (on Thursdays). The historical time series is also weekly updated, and it is composed of IBI BGC best estimates (*from a BGC hindcast run coupled to the IBI physical ocean analysis available).

II.5 Grid

The operational system covers part of the north-east Atlantic Ocean from the Canary Islands (26°N) to Iceland (64°N) and from 20°W to the North Sea (14°E) and the western Mediterranean (10°E), using a curvilinear grid with a horizontal resolution of 1/36°, corresponding to ~2 km for latitudes covered by the IBI domain. The model domain is hereinafter referred as IBI extended domain (**Figure 3**).

However, the IBI-MFC delivers IBI products to end-users of the Copernicus Marine Service over a smaller area, extending from 19°W to 5°E and 26°N to 56°N (referred to as IBI service domain; **Figure 3**), on a regular 1/36° longitude-latitude grid version (constant longitude and latitude step of 0.02778°). The resulting horizontal grid extends to 1081 x 865 grid points.

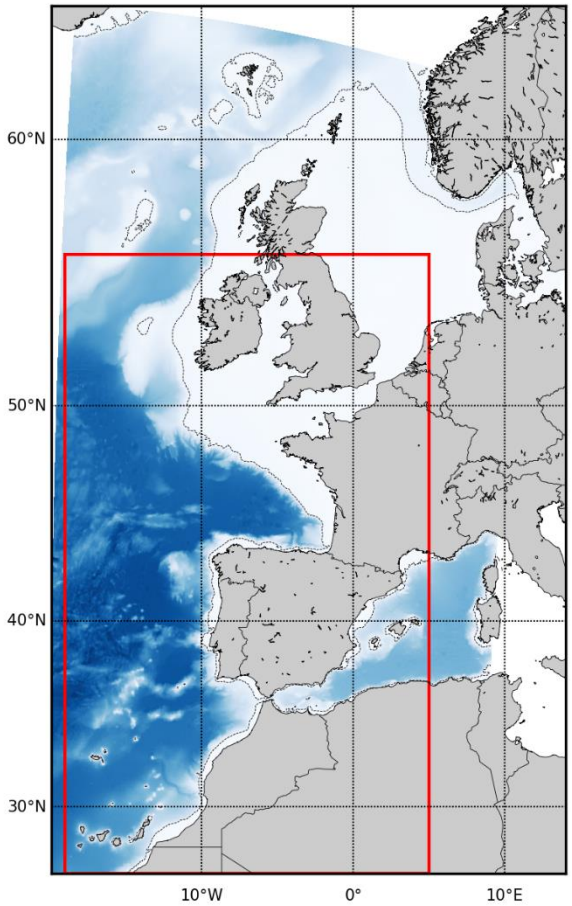


Figure 3. IBI extended domain on the native curvilinear grid and IBI service domain extending from -19°W to 5°E and 26 to 56°N on the regular grid (red rectangle). The blue colour shading represents the bathymetry, and the dashed line is the 200m isobath delimiting the shelf region.

II.6 Vertical Levels

The product is served on the 50 native vertical levels, which are unevenly spaced vertical levels covering the whole column (from 5800 m depth to the surface). The thickness of the layers is around 1-meter resolution near the surface, and up to 400 meters at the bottom of the ocean.

II.7 Processing information

II.7.1 Update Time

The 1-day hindcast + 10-day forecast fields of the daily dataset is updated every Thursday at 18.00UTC.

The monthly file for month-1 is available the 20th of every month.

II.7.2 Temporal extent of analysis and forecast stored on delivery mechanism

The **IBI_ANALYSISFORECAST_BGC_005_004** product is weekly generated an updated.

Each IBI Forecast Cycle produces, at day D, 11 days of ocean information: 10 days of forecast (until D+10d) plus an extra day of hindcast (previous day D-1). These weekly updated hindcast data are delivered as temporary IBI best estimates, being substituted on weekly basis by the IBI hindcast (i.e.: the solution obtained from the hindcast non-assimilative biogeochemical run coupled with the IBI ocean physical analysis), which is kept as final IBI best estimates, making up the historic IBI time series.

The last forecast bulletin is daily updated, being available together with the historic best estimates corresponding to **2 years up to the present time**. That is, the historic catalogue is available in a window of 2 years.

II.7.3 Time averaging

- **cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1D-m &**
  cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1D-m:
  the fields are daily averaged through a 25-hr mean centered at noon (12:00 am).
- **cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1M-m &**
  cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1M-m:

  the fields are monthly means over the calendar month (first to last day of the month).

III FILE FORMAT

The products are stored using the NetCDF format. To know more about the NetCDF format, please follow this link:

What is the format of Copernicus Marine products ? NetCDF

To understand the differences between netCDF and Zarr, please consult this article:

how-to-choose-between-netcdf-and-zarr-format-using-the-toolbox

IV. FILES NOMENCLATURE

Information about nomenclature of files when downloaded can be found in this articles " **How is defined the nomenclature of Copernicus Marine data? | Copernicus Marine Help Center***"*

The file naming and metadata of the downloaded files may differ depending on the chosen download mechanism. The different ways to download the data are explained in the corresponding section. These methodologies allow the download of either the original file format (Get function of the Copernicus Marine Toolbox, and the File browser system), or post-processed files for subsetting approaches (subset function of the Copernicus Marine Toolbox, subsetter page,…).

IV.1 File nomenclature of original files

CMEMS_{fileVersion}_{region}_BIO_NRT_NL_{freqFlag}_{datatype}_{validDate}_{validDate}_R{bulletinDate}_{productType}.nc

Where:

- **region:** a three-letter code for the region, IBI in this case.
- **fileVersion:** vxry, where x, y are the version and release number, respectively.
- **freqFlag:** the frequency of data values in the file (01dav = daily averaged, 01mav = monthly averaged).
- **Datatype (optional):** it is a parameter which inform about the kind of included parameters.
- **validDate**: YYYYMMDD is the valid date of the fields contained in the file. For the monthly dataset, this field is the initial and final date of the month.
- **bulletinDate:** RYYYYMMDD is the bulletin date, i.e.: data production date.
- **productType:** is a two-letter code for the product type (HC01 for hindcast, AN01 for analysis, FC01 for forecast).

Examples:

From the daily dataset request:

CMEMS_v8r1_IBI_BIO_NRT_NL_01dav_20240101_20240101_R20240108_HC01.nc

CMEMS_v8r1_IBI_BIO_NRT_NL_01dav_optics_20240101_20240101_R20240108_HC01.nc

From the monthly dataset request:

CMEMS_v8r1_IBI_BIO_NRT_NL_01mav_20240101_20240131_R20240131_HC01.nc

CMEMS_v8r1_IBI_BIO_NRT_NL_01mav_optics_20240101_20240131_R20240131_HC01.nc

IV.2 Other information: land mask value, missing value

The NetCDF-4 format is used with short integer coding for better compression, using an offset and scale factor as follows:

$$Real_value = (\,Display_value * scale_factor) + add_offset$$

The **missing value** for this product is: **-32767s**

Land mask is equal to **"_FillValue"** (see variable attribute on NetCDF file).

Land values are treated as missing value.

IV.3 File size

DATASET NAME FILE NAME	DIMENSION (no fixed size)
cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1D-m CMEMS_v8r1_IBI_BIO_NRT_NL_01dav_${validDate}_${validDate}_R${bulletinDate}_{productType}.nc	85Mb
cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1M-m CMEMS_v8r1_IBI_BIO_NRT_NL_01mav_${iniDate}_${endDate}_R${bulletinDate}_{productType}.nc	70Mb
cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1D-m CMEMS_v8r1_IBI_BIO_NRT_NL_01dav_optics_${validDate}_${validDate}_R${bulletinDate}_{productType}.nc	8Mb
cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1M-m CMEMS_v8r1_IBI_BIO_NRT_NL_01mav_optics_${validDate}_${validDate}_R${bulletinDate}_{productType}.nc	8Mb
cmems_mod_ibi_bgc_anfc_0.027deg-3D_static IBI-MFC_005_004_coordinates.nc IBI-MFC_005_004_mask_bathy.nc	11Mb 3.0Mb

Table 6. Name and size of files in each dataset (compressed values).

IV.4 Structure of file

Examples of the header of output NetCDF files are inserted in VI

V REFERENCES

Quality Information Document (QUID) :

https://catalogue.marine.copernicus.eu/documents/QUID/CMEMS-IBI-QUID-005-004.pdf

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VI ANNEX

Examples of structure and header of **IBI_ANALYSISFORECAST_BGC_005_004** original files downloaded through the FTP User interfaceCopernicus Marine Toolbox.

cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1D-m & **cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1M-m**
netcdf **CMEMS_v87r1_IBI_BBIOIO_NRT_NL_01dav_20220518_20220518_R20220525_**AN01 { dimensions: time = 1 ; longitude = 865 ; latitude = 1081 ; depth = 50 ; variables: float time(time) ; time:standard_name = "time" ; time:long_name = "time" ; time:units = "hours since 1950-01-01 00:00:00" ; time:calendar = "gregorian" ; time:axis = "T" ; float longitude(longitude) ; longitude:standard_name = "longitude" ; longitude:long_name = "Longitude" ; longitude:units = "degrees_east" ; longitude:unit_long = "Degrees East" ; longitude:step = "0.02777863f" ; longitude:axis = "X" ; float latitude(latitude) ; latitude:standard_name = "latitude" ; latitude:long_name = "Latitude" ; latitude:units = "degrees_north" ; latitude:unit_long = "Degrees North" ; latitude:step = "0.02777863f" ; latitude:axis = "Y" ; float time(time) ; time:axis = "T" ; time:calendar = "gregorian" ; time:standard_name = "time" ; time:units = "hours since 1950-01-01 00:00:00" ; time:long_name = "time" ; float longitude(longitude) ; longitude:standard_name = "longitude" ; longitude:long_name = "Longitude" ; longitude:units = "degrees_east" ; longitude:axis = "X" ; longitude:unit_long = "Degrees East" ;

```
        longitude:step = "0.02777863f" ;
    float latitude(latitude) ;
        latitude:standard_name = "latitude" ;
        latitude:long_name = "Latitude" ;
        latitude:units = "degrees_north" ;
        latitude:axis = "Y" ;
        latitude:unit_long = "Degrees North" ;
        latitude:step = "0.02777863f" ;
    float depth(depth) ;
        depth:long_name = "Depth" ;
        depth:units = "m" ;
        depth:axis = "Z" ;
        depth:positive = "down" ;
        depth:unit_long = "Meters" ;
        depth:standard_name = "depth" ;
    float depth(depth) ;
        depth:standard_name = "depth" ;
        depth:long_name = "Depth" ;
        depth:units = "m" ;
        depth:unit_long = "Meters" ;
        depth:positive = "down" ;
        depth:axis = "Z" ;
    short chl(time, depth, latitude, longitude) ;
        chl:add_offset = 0.f ;
        chl:scale_factor = 0.01f ;
        chl:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
        chl:long_name = "Mass Concentration of Chlorophyll in Sea Water" ;
        chl:units = "mg.m-3" ;
        chl:unit_long = "milligrams of chlorophyll per cubic meter" ;
        chl:valid_min = 0s ;
        chl:valid_max = 20000s ;
        chl:easting = "longitude" ;
        chl:northing = "latitude" ;
        chl:longitude_min = “-19.0f”  ;
        chl:longitude_max = “5.000736f” ;
        chl:latitude_min = “26.0f” ;
        chl:latitude_max = “56.00092f” ;
        chl:z_min = "0.49402f" ;
        chl:z_max = "5727.917f" ;
        chl:_FillValue = -32767s ;
    short zeu(time, latitude, longitude) ;
        zeu:add_offset = 0.f ;
        zeu:scale_factor = 0.1f ;
        zeu:standard_name = "euphotic_zone_depth" ;
        zeu:long_name = "Euphotic Zone Depth" ;
        zeu:units = "m" ;
        zeu:unit_long = "meters" ;
        zeu:valid_min = 0s ;
        zeu:valid_max = 2000s ;
```

```
        zeu:easting = "longitude" ;
        zeu:northing = "latitude" ;
        zeu:longitude_min = “-19.0f”  ;
        zeu:longitude_max = “5.000736f” ;
        zeu:latitude_min = “26.0f” ;
        zeu:latitude_max = “56.00092f” ;
        zeu:z_min = "0.494025f" ;
        zeu:z_max = "0.494025f" ;
        zeu:_FillValue = -32767s ;
    short fe(time, depth, latitude, longitude) ;
        fe:add_offset = 0.f ;
        fe:scale_factor = 0.0001f ;
        fe:standard_name = "mole_concentration_of_dissolved_iron_in_sea_water" ;
        fe:long_name = "Mole Concentration of Iron in Sea Water" ;
        fe:units = "mmol.m-3" ;
        fe:unit_long = "millimoles of Iron per cubic meter" ;
        fe:valid_min = 0s ;
        fe:valid_max = 1000s ;
        fe:easting = "longitude" ;
        fe:northing = "latitude" ;
        fe:longitude_min = “-19.0f”  ;
        fe:longitude_max = “5.000736f” ;
        fe:latitude_min = “26.0f” ;
        fe:latitude_max = “56.00092f” ;
        fe:z_min = "0.49402f" ;
        fe:z_max = "5727.917f" ;
        fe:_FillValue = -32767s ;
    short nh4(time, depth, latitude, longitude) ;
        nh4:add_offset = 0.f ;
        nh4:scale_factor = 0.1f ;
        nh4:standard_name = "mole_concentration_of_ammonium_in_sea_water" ;
        nh4:long_name = "Mole Concentration of Ammonium in Sea Water" ;
        nh4:units = "mmol.m-3" ;
        nh4:unit_long = "millimoles of Ammonium per cubic meter" ;
        nh4:valid_min = 0s ;
        nh4:valid_max = 3200s ;
        nh4:easting = "longitude" ;
        nh4:northing = "latitude" ;
        nh4:longitude_min = “-19.0f”  ;
        nh4:longitude_max = “5.000736f” ;
        nh4:latitude_min = “26.0f” ;
        nh4:latitude_max = “56.00092f” ;
        nh4:z_min = "0.49402f" ;
        nh4:z_max = "5727.917f" ;
        nh4:_FillValue = -32767s ;
    short no3(time, depth, latitude, longitude) ;
        no3:add_offset = 300.f ;
        no3:scale_factor = 0.1f ;
        no3:standard_name = "mole_concentration_of_nitrate_in_sea_water" ;
```

```
            no3:long_name = "Mole Concentration of Nitrate in Sea Water" ;
            no3:units = "mmol.m-3" ;
            no3:unit_long = "millimoles of Nitrate per cubic meter" ;
            no3:valid_min = -3000s ;
            no3:valid_max = 3000s ;
            no3:easting = "longitude" ;
            no3:northing = "latitude" ;
            no3:longitude_min = “-19.0f”  ;
            no3:longitude_max = “5.000736f” ;
            no3:latitude_min = “26.0f” ;
            no3:latitude_max = “56.00092f” ;
            no3:z_min = "0.49402f" ;
            no3:z_max = "5727.917f" ;
            no3:_FillValue = -32767s ;
      short o2(time, depth, latitude, longitude) ;
            o2:add_offset = 0.f ;
            o2:scale_factor = 1.f ;
            o2:standard_name =
"mole_concentration_of_dissolved_molecular_oxygen_in_sea_water" ;
            o2:long_name = "Mole Concentration of Dissolved Oxygen in Sea Water" ;
            o2:units = "mmol.m-3" ;
            o2:unit_long = "millimoles of Oxygen per cubic meter" ;
            o2:valid_min = 0s ;
            o2:valid_max = 3200s ;
            o2:easting = "longitude" ;
            o2:northing = "latitude" ;
            o2:longitude_min = “-19.0f”  ;
            o2:longitude_max = “5.000736f” ;
            o2:latitude_min = “26.0f” ;
            o2:latitude_max = “56.00092f” ;
            o2:z_min = "0.49402f" ;
            o2:z_max = "5727.917f" ;
            o2:_FillValue = -32767s ;
      short phyc(time, depth, latitude, longitude) ;
            phyc:add_offset = 300.f ;
            phyc:scale_factor = 0.1f ;
            phyc:standard_name =
"mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water" ;
            phyc:long_name = "Mole Concentration of Phytoplankton Expressed as Carbon in Sea
Water" ;
            phyc:units = "mmol.m-3" ;
            phyc:unit_long = "millimoles of Phytoplankton Carbon per cubic meter" ;
            phyc:valid_min = -3000s ;
            phyc:valid_max = 3000s ;
            phyc:easting = "longitude" ;
            phyc:northing = "latitude" ;
            phyc:longitude_min = “-19.0f”  ;
            phyc:longitude_max = “5.000736f” ;
            phyc:latitude_min = “26.0f” ;
```

```
            phyc:latitude_max = “56.00092f” ;
            phyc:z_min = "0.49402f" ;
            phyc:z_max = "5727.917f" ;
            phyc:_FillValue = -32767s ;
      short po4(time, depth, latitude, longitude) ;
            po4:add_offset = 20.f ;
            po4:scale_factor = 0.01f ;
            po4:standard_name = "mole_concentration_of_phosphate_in_sea_water" ;
            po4:long_name = "Mole Concentration of Phosphate in Sea Water" ;
            po4:units = "mmol.m-3" ;
            po4:unit_long = "millimoles of Phosphate per cubic meter" ;
            po4:valid_min = -2000s ;
            po4:valid_max = 3000s ;
            po4:easting = "longitude" ;
            po4:northing = "latitude" ;
            po4:longitude_min = “-19.0f”  ;
            po4:longitude_max = “5.000736f” ;
            po4:latitude_min = “26.0f” ;
            po4:latitude_max = “56.00092f” ;
            po4:z_min = "0.49402f" ;
            po4:z_max = "5727.917f" ;
            po4:_FillValue = -32767s ;
      short nppv(time, depth, latitude, longitude) ;
            nppv:add_offset = 0.f ;
            nppv:scale_factor = 1.f ;
            nppv:standard_name =
"net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water" ;
            nppv:long_name = "Net Primary Productivity of Carbon" ;
            nppv:units = "mg.m-3.day-1" ;
            nppv:unit_long = "milligrams of Carbon per cubic meter per day" ;
            nppv:valid_min = 0s ;
            nppv:valid_max = 10000s ;
            nppv:easting = "longitude" ;
            nppv:northing = "latitude" ;
            nppv:longitude_min = “-19.0f”  ;
            nppv:longitude_max = “5.000736f” ;
            nppv:latitude_min = “26.0f” ;
            nppv:latitude_max = “56.00092f” ;
            nppv:z_min = "0.49402f" ;
            nppv:z_max = "5727.917f" ;
            nppv:_FillValue = -32767s ;
      short si(time, depth, latitude, longitude) ;
            si:add_offset = 0.f ;
            si:scale_factor = 1.f ;
            si:standard_name = "mole_concentration_of_silicate_in_sea_water" ;
            si:long_name = "Mole Concentration of Silicate in Sea Water" ;
            si:units = "mmol.m-3" ;
            si:unit_long = "millimoles of Silicate per cubic meter" ;
            si:valid_min = 0s ;
```

```
            si:valid_max = 2500s ;
            si:easting = "longitude" ;
            si:northing = "latitude" ;
            si:longitude_min = “-19.0f”  ;
            si:longitude_max = “5.000736f” ;
            si:latitude_min = “26.0f” ;
            si:latitude_max = “56.00092f” ;
            si:z_min = "0.49402f" ;
            si:z_max = "5727.917f" ;
            si:_FillValue = -32767s ;
      short dissic(time, depth, latitude, longitude) ;
            dissic:add_offset = 10.f ;
            dissic:scale_factor = 0.01f ;
            dissic:standard_name =
"mole_concentration_of_dissolved_inorganic_carbon_in_sea_water" ;
            dissic:long_name = "Dissolved Inorganic Carbon" ;
            dissic:units = "mol.m-3" ;
            dissic:unit_long = "moles of Carbon per cubic meter" ;
            dissic:valid_min = -1000s ;
            dissic:valid_max = -200s ;
            dissic:easting = "longitude" ;
            dissic:northing = "latitude" ;
            dissic:longitude_min = “-19.0f”  ;
            dissic:longitude_max = “5.000736f” ;
            dissic:latitude_min = “26.0f” ;
            dissic:latitude_max = “56.00092f” ;
            dissic:z_min = "0.49402f" ;
            dissic:z_max = "5727.917f" ;
            dissic:_FillValue = -32767s ;
      short ph(time, depth, latitude, longitude) ;
            ph:add_offset = 6.f ;
            ph:scale_factor = 0.0004f ;
            ph:standard_name = "sea_water_ph_reported_on_total_scale" ;
            ph:long_name = "sea water ph reported on total scale" ;
            ph:units = "1" ;
            ph:unit_long = "1" ;
            ph:valid_min = -2500s ;
            ph:valid_max = 7500s ;
            ph:easting = "longitude" ;
            ph:northing = "latitude" ;
            ph:longitude_min = “-19.0f”  ;
            ph:longitude_max = “5.000736f” ;
            ph:latitude_min = “26.0f” ;
            ph:latitude_max = “56.00092f” ;
            ph:z_min = "0.49402f" ;
            ph:z_max = "5727.917f" ;
            ph:_FillValue = -32767s ;
      short spco2(time, latitude, longitude) ;
            spco2:add_offset = 500.f ;
```

```
            spco2:scale_factor = 0.02f ;
            spco2:standard_name = "surface_partial_pressure_of_carbon_dioxide_in_sea_water" ;
            spco2:long_name = "surface partial pressure of carbon dioxide in sea water" ;
            spco2:units = "Pa" ;
            spco2:unit_long = "Pascal" ;
            spco2:valid_min = -24900s ;
            spco2:valid_max = 10000s ;
            spco2:easting = "longitude" ;
            spco2:northing = "latitude" ;
            spco2:longitude_min = “-19.0f”  ;
            spco2:longitude_max = “5.000736f” ;
            spco2:latitude_min = “26.0f” ;
            spco2:latitude_max = “56.00092f” ;
            spco2:z_min = "0.494025f" ;
            spco2:z_max = "0.494025f" ;
            spco2:_FillValue = -32767s ;
      short zooc(time, depth, latitude, longitude) ;
            zooc:add_offset = 10.f ;
            zooc:scale_factor = 0.01f ;
            zooc:standard_name =
"mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water" ;
            zooc:long_name = "Mole Concentration of Zooplankton Expressed as Carbon in Sea Water"
;
            zooc:units = "mmol.m-3" ;
            zooc:unit_long = "millimoles of Zooplankton per cubic meter" ;
            zooc:valid_min = -1000s ;
            zooc:valid_max = 9000s ;
            zooc:easting = "longitude" ;
            zooc:northing = "latitude" ;
            zooc:longitude_min = “-19.0f”  ;
            zooc:longitude_max = “5.000736f” ;
            zooc:latitude_min = “26.0f” ;
            zooc:latitude_max = “56.00092f” ;
            zooc:z_min = "0.49402f" ;
            zooc:z_max = "5727.917f" ;
            zooc:_FillValue = -32767s ;       short chl(time, depth, latitude, longitude) ;

            chl:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
            chl:long_name = "Mass Concentration of Chlorophyll in Sea Water" ;
            chl:units = "mg.m-3" ;
            chl:add_offset = 0.f ;
            chl:scale_factor = 0.01f ;
            chl:_FillValue = -32767s ;
            chl:unit_long = "milligrams of chlorophyll per cubic meter" ;
            chl:valid_max = 20000s ;
            chl:valid_min = 0s ;
      short zeu(time, latitude, longitude) ;
            zeu:standard_name = "euphotic_zone_depth" ;
            zeu:long_name = "Euphotic Zone Depth" ;
```

```
            zeu:units = "m" ;
            zeu:add_offset = 0.f ;
            zeu:scale_factor = 0.1f ;
            zeu:_FillValue = -32767s ;
            zeu:unit_long = "meters" ;
            zeu:valid_max = 2000s ;
            zeu:valid_min = 0s ;
      short fe(time, depth, latitude, longitude) ;
            fe:standard_name = "mole_concentration_of_dissolved_iron_in_sea_water" ;
            fe:long_name = "Mole Concentration of Iron in Sea Water" ;
            fe:units = "mmol.m-3" ;
            fe:add_offset = 0.f ;
            fe:scale_factor = 0.0001f ;
            fe:_FillValue = -32767s ;
            fe:unit_long = "millimoles of Iron per cubic meter" ;
            fe:valid_max = 1000s ;
            fe:valid_min = 0s ;
      short nh4(time, depth, latitude, longitude) ;
            nh4:standard_name = "mole_concentration_of_ammonium_in_sea_water" ;
            nh4:long_name = "Mole Concentration of Ammonium in Sea Water" ;
            nh4:units = "mmol.m-3" ;
            nh4:add_offset = 0.f ;
            nh4:scale_factor = 0.1f ;
            nh4:_FillValue = -32767s ;
            nh4:unit_long = "millimoles of Ammonium per cubic meter" ;
            nh4:valid_max = 3200s ;
            nh4:valid_min = 0s ;
      short no3(time, depth, latitude, longitude) ;
            no3:standard_name = "mole_concentration_of_nitrate_in_sea_water" ;
            no3:long_name = "Mole Concentration of Nitrate in Sea Water" ;
            no3:units = "mmol.m-3" ;
            no3:add_offset = 300.f ;
            no3:scale_factor = 0.1f ;
            no3:_FillValue = -32767s ;
            no3:unit_long = "millimoles of Nitrate per cubic meter" ;
            no3:valid_max = 3000s ;
            no3:valid_min = -3000s ;
      short o2(time, depth, latitude, longitude) ;
            o2:standard_name =
"mole_concentration_of_dissolved_molecular_oxygen_in_sea_water" ;
            o2:long_name = "Mole Concentration of Dissolved Oxygen in Sea Water" ;
            o2:units = "mmol.m-3" ;
            o2:_FillValue = -32767s ;
            o2:unit_long = "millimoles of Oxygen per cubic meter" ;
            o2:valid_max = 3200s ;
            o2:valid_min = 0s ;
            o2:add_offset = 0.f ;
            o2:scale_factor = 1.f ;
      short phyc(time, depth, latitude, longitude) ;
```

```
            phyc:standard_name =
"mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water" ;
            phyc:long_name = "Mole Concentration of Phytoplankton Expressed as Carbon in Sea
Water" ;
            phyc:units = "mmol.m-3" ;
            phyc:add_offset = 300.f ;
            phyc:scale_factor = 0.1f ;
            phyc:_FillValue = -32767s ;
            phyc:unit_long = "millimoles of Phytoplankton Carbon per cubic meter" ;
            phyc:valid_max = 3000s ;
            phyc:valid_min = -3000s ;
      short po4(time, depth, latitude, longitude) ;
            po4:standard_name = "mole_concentration_of_phosphate_in_sea_water" ;
            po4:long_name = "Mole Concentration of Phosphate in Sea Water" ;
            po4:units = "mmol.m-3" ;
            po4:add_offset = 20.f ;
            po4:scale_factor = 0.01f ;
            po4:_FillValue = -32767s ;
            po4:unit_long = "millimoles of Phosphate per cubic meter" ;
            po4:valid_max = 3000s ;
            po4:valid_min = -2000s ;
      short nppv(time, depth, latitude, longitude) ;
            nppv:standard_name =
"net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water" ;
            nppv:long_name = "Net Primary Productivity of Carbon" ;
            nppv:units = "mg.m-3.day-1" ;
            nppv:_FillValue = -32767s ;
            nppv:unit_long = "milligrams of Carbon per cubic meter per day" ;
            nppv:valid_max = 10000s ;
            nppv:valid_min = 0s ;
            nppv:add_offset = 0.f ;
            nppv:scale_factor = 1.f ;
      short si(time, depth, latitude, longitude) ;
            si:standard_name = "mole_concentration_of_silicate_in_sea_water" ;
            si:long_name = "Mole Concentration of Silicate in Sea Water" ;
            si:units = "mmol.m-3" ;
            si:_FillValue = -32767s ;
            si:unit_long = "millimoles of Silicate per cubic meter" ;
            si:valid_max = 2500s ;
            si:valid_min = 0s ;
            si:add_offset = 0.f ;
            si:scale_factor = 1.f ;
      short spco2(time, latitude, longitude) ;
            spco2:standard_name = "surface_partial_pressure_of_carbon_dioxide_in_sea_water" ;
            spco2:long_name = "surface partial pressure of carbon dioxide in sea water" ;
            spco2:units = "Pa" ;
            spco2:add_offset = 500.f ;
            spco2:scale_factor = 0.02f ;
            spco2:_FillValue = -32767s ;
```

```
            spco2:unit_long = "Pascal" ;
            spco2:valid_max = 10000s ;
            spco2:valid_min = -24900s ;
      short dissic(time, depth, latitude, longitude) ;
            dissic:standard_name =
"mole_concentration_of_dissolved_inorganic_carbon_in_sea_water" ;
            dissic:long_name = "Dissolved Inorganic Carbon" ;
            dissic:units = "mol.m-3" ;
            dissic:add_offset = 10.f ;
            dissic:scale_factor = 0.01f ;
            dissic:_FillValue = -32767s ;
            dissic:unit_long = "moles of Carbon per cubic meter" ;
            dissic:valid_max = -200s ;
            dissic:valid_min = -1000s ;
      short ph(time, depth, latitude, longitude) ;
            ph:standard_name = "sea_water_ph_reported_on_total_scale" ;
            ph:long_name = "sea water ph reported on total scale" ;
            ph:units = "1" ;
            ph:add_offset = 6.f ;
            ph:scale_factor = 0.0004f ;
            ph:_FillValue = -32767s ;
            ph:unit_long = "1" ;
            ph:valid_max = 7500s ;
            ph:valid_min = 1500s ;
      short zooc(time, depth, latitude, longitude) ;
            zooc:standard_name =
"mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water " ;
            zooc:long_name = "Mole Concentration of Zooplankton Expressed as Carbon in Sea Water"
;
            zooc:units = "mmol.m-3" ;
            zooc:add_offset = 10.f ;
            zooc:scale_factor = 0.01f ;
            zooc:_FillValue = -32767s ;
            zooc:unit_long = "millimoles of Zooplankton Carbon per cubic meter" ;
            zooc:valid_min = -1000s ;
            zooc:valid_max = 9000s ;

// global attributes:
            :Conventions = "CF-1.8" ;
            :title = "Biogeochemical 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :source = "NEMO3.6-PISCES3.6" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "20220517" ;
            :forecast_range = "02-day_forecast" ;
            :forecast_type = "forecast" ;
            :institution = "Nologin (Spain)" ;
            :references = "http://marine.copernicus.eu/" ;
            :contact = "https://marine.copernicus.eu/contact" ;
```

```
            :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
            :comment = "" ;
}// global attributes:
```

Metadata for monthly files is exactly the same (just change the tittle global attribute :

netcdf **CMEMS_v8r1_IBI_BIO_NRT_NL_01mav_20220401_20220430_R20220430_AN01** {

```
…
// global attributes:
…            :Conventions = "CF-1.0" ;

            :title = "Biogeochemical 3D monthly mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
…            :source = "NEMO3.6-PISCESv2" ;
            :institution = "Nologin (Spain) - Mercator Ocean (France)" ;
            :title = "Biogeochemical 3D daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :easting = "longitude" ;
            :northing = "latitude" ;
            :references = "http://marine.copernicus.eu/" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "20220518" ;
            :forecast_range = "01-day_analysis" ;
            :forecast_type = "analysis" ;
            :bulletin_date = "2022-05-25" ;
            :bulletin_type = "Operational" ;
            :longitude_min = "-19.0f" ;
            :longitude_max = "5.000736f" ;
            :latitude_min = "26.0f" ;
            :latitude_max = "56.00092f" ;
            :z_min = "0.49402f" ;
            :z_max = "5727.917f" ;
            :contact = "mailto: servicedesk.cmems@mercator-ocean.eu" ;
}
```

cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1D-m &
cmems_mod_ibi_bgc_optics_anfc_0.027deg_P1M-m:
cmems_mod_ibi_bgc_anfc_0.027deg-3D_P1M-m

netcdf
CMEMS_v87r1_IBI_BBIOIO_NRT_NL_01dmav_optics_20220401_20220430_R20220430_AN01 {

```
dimensions:
      time = 1 ;
      longitude = 865 ;
      latitude = 1081 ;
      depth = 50 ;
variables:
float time(time) ;
            time:standard_name = "time" ;
            time:long_name = "time" ;
```

```
            time:units = "hours since 1950-01-01 00:00:00" ;
            time:calendar = "gregorian" ;
            time:axis = "T" ;
      float longitude(longitude) ;
            longitude:standard_name = "longitude" ;
            longitude:long_name = "Longitude" ;
            longitude:units = "degrees_east" ;
            longitude:unit_long = "Degrees East" ;
            longitude:step = "0.02777863f" ;
            longitude:axis = "X" ;
      float latitude(latitude) ;
            latitude:standard_name = "latitude" ;
            latitude:long_name = "Latitude" ;
            latitude:units = "degrees_north" ;
            latitude:unit_long = "Degrees North" ;
            latitude:step = "0.02777863f" ;
            latitude:axis = "Y" ;
      float depth(depth) ;
            depth:standard_name = "depth" ;
            depth:long_name = "Depth" ;
            depth:units = "m" ;
            depth:unit_long = "Meters" ;
            depth:positive = "down" ;
            depth:axis = "Z" ;dimensions:
      time = 1 ;
      depth = 50 ;
      latitude = 1081 ;
      longitude = 865 ;
      short kd(time, depth, latitude, longitude) ;
            kd:add_offset = 0.f ;
            kd:scale_factor = 0.0008f ;
            kd:standard_name =
"volume_attenuation_coefficient_of_downwelling_radiative_flux_in_sea_water" ;
            kd:long_name = "diffuse attenuation coeff of light in sea water, averaged over the 400-500
nm range" ;
            kd:units = "m-1" ;
            kd:unit_long = "fractional change of radiative flux per unit path length" ;
            kd:valid_min = 0s ;
            kd:valid_max = 32500s ;
            kd:easting = "longitude" ;
            kd:northing = "latitude" ;
            kd:longitude_min = “-19.0f”  ;
            kd:longitude_max = “5.000736f” ;
            kd:latitude_min = “26.0f” ;
            kd:latitude_max = “56.00092f” ;
            kd:z_min = "0.49402f" ;
            kd:z_max = "5727.917f" ;
            kd:_FillValue = -32767s ;
```

```
// global attributes:
        :Conventions = "CF-1.8" ;
        :title = "Biogeochemical (optics) daily mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
        :source = "NEMO3.6-PISCES3.6" ;
        :domain_name = "IBI36" ;
        :field_type = "mean" ;
        :field_date = "20220517" ;
        :forecast_range = "02-day_forecast" ;
        :forecast_type = "forecast" ;
        :institution = "Nologin (Spain)" ;
        :references = "http://marine.copernicus.eu/" ;
        :contact = "https://marine.copernicus.eu/contact" ;
        :licence = "https://marine.copernicus.eu/user-corner/service-commitments-and-licence" ;
        :comment = "" ;
}
```

Metadata for monthly files is exactly the same (just change the tittle global attribute :

```
netcdf CMEMS_v8r1_IBI_BIO_NRT_NL_01mav_optics_20220401_20220430_R20220430_AN01 {
…
// global attributes:
…
        :title = "Biogeochemical (optics) monthly mean fields for the Iberia-Biscay-Ireland (IBI)
region" ;
…variables:
    short chl(time, depth, latitude, longitude) ;
        chl:standard_name = "mass_concentration_of_chlorophyll_a_in_sea_water" ;
        chl:long_name = "Mass Concentration of Chlorophyll in Sea Water" ;
        chl:units = "mg.m-3" ;
        chl:add_offset = 0.f ;
        chl:scale_factor = 0.01f ;
        chl:_FillValue = -32767s ;
        chl:unit_long = "milligrams of chlorophyll per cubic meter" ;
        chl:valid_max = 20000s ;
        chl:valid_min = 0s ;
    float depth(depth) ;
        depth:long_name = "Depth" ;
        depth:units = "m" ;
        depth:axis = "Z" ;
        depth:positive = "down" ;
        depth:unit_long = "Meters" ;
        depth:standard_name = "depth" ;
    short dissic(time, depth, latitude, longitude) ;
        dissic:standard_name =
"mole_concentration_of_dissolved_inorganic_carbon_in_sea_water" ;
        dissic:long_name = "Dissolved Inorganic Carbon" ;
        dissic:units = "mol.m-3" ;
        dissic:add_offset = 10.f ;
        dissic:scale_factor = 0.01f ;
        dissic:_FillValue = -32767s ;
```

```
        dissic:unit_long = "moles of Carbon per cubic meter" ;
        dissic:valid_max = -200s ;
        dissic:valid_min = -1000s ;
    short fe(time, depth, latitude, longitude) ;
        fe:standard_name = "mole_concentration_of_dissolved_iron_in_sea_water" ;
        fe:long_name = "Mole Concentration of Iron in Sea Water" ;
        fe:units = "mmol.m-3" ;
        fe:add_offset = 0.f ;
        fe:scale_factor = 0.0001f ;
        fe:_FillValue = -32767s ;
        fe:unit_long = "millimoles of Iron per cubic meter" ;
        fe:valid_max = 1000s ;
        fe:valid_min = 0s ;
    float latitude(latitude) ;
        latitude:standard_name = "latitude" ;
        latitude:long_name = "Latitude" ;
        latitude:units = "degrees_north" ;
        latitude:axis = "Y" ;
        latitude:unit_long = "Degrees North" ;
        latitude:step = "0.02777863f" ;
    float longitude(longitude) ;
        longitude:standard_name = "longitude" ;
        longitude:long_name = "Longitude" ;
        longitude:units = "degrees_east" ;
        longitude:axis = "X" ;
        longitude:unit_long = "Degrees East" ;
        longitude:step = "0.02777863f" ;
    short nh4(time, depth, latitude, longitude) ;
        nh4:standard_name = "mole_concentration_of_ammonium_in_sea_water" ;
        nh4:long_name = "Mole Concentration of Ammonium in Sea Water" ;
        nh4:units = "mmol.m-3" ;
        nh4:add_offset = 0.f ;
        nh4:scale_factor = 0.1f ;
        nh4:_FillValue = -32767s ;
        nh4:unit_long = "millimoles of Ammonium per cubic meter" ;
        nh4:valid_max = 3200s ;
        nh4:valid_min = 0s ;
    short no3(time, depth, latitude, longitude) ;
        no3:standard_name = "mole_concentration_of_nitrate_in_sea_water" ;
        no3:long_name = "Mole Concentration of Nitrate in Sea Water" ;
        no3:units = "mmol.m-3" ;
        no3:add_offset = 300.f ;
        no3:scale_factor = 0.1f ;
        no3:_FillValue = -32767s ;
        no3:unit_long = "millimoles of Nitrate per cubic meter" ;
        no3:valid_max = 3000s ;
        no3:valid_min = -3000s ;
    short nppv(time, depth, latitude, longitude) ;
```

```
            nppv:standard_name =
"net_primary_production_of_biomass_expressed_as_carbon_per_unit_volume_in_sea_water" ;
            nppv:long_name = "Net Primary Productivity of Carbon" ;
            nppv:units = "mg.m-3.day-1" ;
            nppv:_FillValue = -32767s ;
            nppv:unit_long = "milligrams of Carbon per cubic meter per day" ;
            nppv:valid_max = 10000s ;
            nppv:valid_min = 0s ;
            nppv:add_offset = 0.f ;
            nppv:scale_factor = 1.f ;
      short o2(time, depth, latitude, longitude) ;
            o2:standard_name =
"mole_concentration_of_dissolved_molecular_oxygen_in_sea_water" ;
            o2:long_name = "Mole Concentration of Dissolved Oxygen in Sea Water" ;
            o2:units = "mmol.m-3" ;
            o2:_FillValue = -32767s ;
            o2:unit_long = "millimoles of Oxygen per cubic meter" ;
            o2:valid_max = 3200s ;
            o2:valid_min = 0s ;
            o2:add_offset = 0.f ;
            o2:scale_factor = 1.f ;
      short ph(time, depth, latitude, longitude) ;
            ph:standard_name = "sea_water_ph_reported_on_total_scale" ;
            ph:long_name = "sea water ph reported on total scale" ;
            ph:units = "1" ;
            ph:add_offset = 6.f ;
            ph:scale_factor = 0.0004f ;
            ph:_FillValue = -32767s ;
            ph:unit_long = "1" ;
            ph:valid_max = 7500s ;
            ph:valid_min = 1500s ;
      short phyc(time, depth, latitude, longitude) ;
            phyc:standard_name =
"mole_concentration_of_phytoplankton_expressed_as_carbon_in_sea_water" ;
            phyc:long_name = "Mole Concentration of Phytoplankton Expressed as Carbon in Sea
Water" ;
            phyc:units = "mmol.m-3" ;
            phyc:add_offset = 300.f ;
            phyc:scale_factor = 0.1f ;
            phyc:_FillValue = -32767s ;
            phyc:unit_long = "millimoles of Phytoplankton Carbon per cubic meter" ;
            phyc:valid_max = 3000s ;
            phyc:valid_min = -3000s ;
      short po4(time, depth, latitude, longitude) ;
            po4:standard_name = "mole_concentration_of_phosphate_in_sea_water" ;
            po4:long_name = "Mole Concentration of Phosphate in Sea Water" ;
            po4:units = "mmol.m-3" ;
            po4:add_offset = 20.f ;
            po4:scale_factor = 0.01f ;
```

```
            po4:_FillValue = -32767s ;
            po4:unit_long = "millimoles of Phosphate per cubic meter" ;
            po4:valid_max = 3000s ;
            po4:valid_min = -2000s ;
      short si(time, depth, latitude, longitude) ;
            si:standard_name = "mole_concentration_of_silicate_in_sea_water" ;
            si:long_name = "Mole Concentration of Silicate in Sea Water" ;
            si:units = "mmol.m-3" ;
            si:_FillValue = -32767s ;
            si:unit_long = "millimoles of Silicate per cubic meter" ;
            si:valid_max = 2500s ;
            si:valid_min = 0s ;
            si:add_offset = 0.f ;
            si:scale_factor = 1.f ;
      short spco2(time, latitude, longitude) ;
            spco2:standard_name = "surface_partial_pressure_of_carbon_dioxide_in_sea_water" ;
            spco2:long_name = "surface partial pressure of carbon dioxide in sea water" ;
            spco2:units = "Pa" ;
            spco2:add_offset = 500.f ;
            spco2:scale_factor = 0.02f ;
            spco2:_FillValue = -32767s ;
            spco2:unit_long = "Pascal" ;
            spco2:valid_max = 10000s ;
            spco2:valid_min = -24900s ;
      float time(time) ;
            time:axis = "T" ;
            time:calendar = "gregorian" ;
            time:standard_name = "time" ;
            time:units = "hours since 1950-01-01 00:00:00" ;
            time:long_name = "time" ;
      short zeu(time, latitude, longitude) ;
            zeu:standard_name = "euphotic_zone_depth" ;
            zeu:long_name = "Euphotic Zone Depth" ;
            zeu:units = "m" ;
            zeu:add_offset = 0.f ;
            zeu:scale_factor = 0.1f ;
            zeu:_FillValue = -32767s ;
            zeu:unit_long = "meters" ;
            zeu:valid_max = 2000s ;
            zeu:valid_min = 0s ;
      short zooc(time, depth, latitude, longitude) ;
            zooc:standard_name =
"mole_concentration_of_zooplankton_expressed_as_carbon_in_sea_water " ;
            zooc:long_name = "Mole Concentration of Zooplankton Expressed as Carbon in Sea Water"
;
            zooc:units = "mmol.m-3" ;
            zooc:add_offset = 10.f ;
            zooc:scale_factor = 0.01f ;
            zooc:_FillValue = -32767s ;
```

```
            zooc:unit_long = "millimoles of Zooplankton Carbon per cubic meter" ;
            zooc:valid_min = -1000s ;
            zooc:valid_max = 9000s ;

// global attributes:
            :Conventions = "CF-1.0" ;
            :source = "NEMO3.6-PISCESv2" ;
            :institution = "Nologin (Spain) - Mercator Ocean (France)" ;
            :title = "Biogeochemical 3D monthly mean fields for the Iberia-Biscay-Ireland (IBI) region" ;
            :easting = "longitude" ;
            :northing = "latitude" ;
            :references = "http://marine.copernicus.eu/" ;
            :domain_name = "IBI36" ;
            :field_type = "mean" ;
            :field_date = "20220416" ;
            :forecast_range = "01-day_analysis" ;
            :forecast_type = "analysis" ;
            :bulletin_date = "2022-04-30" ;
            :bulletin_type = "Operational" ;
            :longitude_min = "-19.0f" ;
            :longitude_max = "5.000736f" ;
            :latitude_min = "26.0f" ;
            :latitude_max = "56.00092f" ;
            :z_min = "0.49402f" ;
            :z_max = "5727.917f" ;
            :contact = "mailto: servicedesk.cmems@mercator-ocean.eu" ;
}
```

cmems_mod_ibi_bgc_anfc_0.027deg-3D_static
This dataset only contains 2 files:
IBI-MFC_005_004_mask_bathy.nc
IBI-MFC_005_004_coordinates.nc

```
netcdf IBI-MFC_005_004_mask_bathy {
    dimensions:
        latitude = 1081 ;
        longitude = 865 ;
        depth = 50 ;
    variables:
        float deptho_lev(latitude, longitude) ;
            deptho_lev:long_name = "Model level number at sea floor" ;
            deptho_lev:standard_name = "model_level_number_at_sea_floor" ;
            deptho_lev:units = "1" ;
            deptho_lev:_FillValue = -32767.f ;
        float longitude(longitude) ;
            longitude:axis = "X" ;
            longitude:standard_name = "longitude" ;
            longitude:units = "degrees_east" ;
```

```
            longitude:step = "0.02777863f" ;
        float latitude(latitude) ;
            latitude:axis = "Y" ;
            latitude:standard_name = "latitude" ;
            latitude:units = "degrees_north" ;
            latitude:step = "0.02777863f" ;
        byte mask(depth, latitude, longitude) ;
            mask:long_name = "Land-sea mask: 1 = sea ; 0 = land" ;
            mask:standard_name = "sea_binary_mask" ;
            mask:units = "1" ;
        float depth(depth) ;
            depth:standard_name = "depth" ;
            depth:units = "m" ;
            depth:positive = "down" ;
            depth:axis = "Z" ;
        float deptho(latitude, longitude) ;
            deptho:_FillValue = -32767.f ;
            deptho:long_name = "Bathymetry" ;
            deptho:standard_name = "sea_floor_depth_below_geoid" ;
            deptho:units = "m" ;

    // global attributes:
            :source = "IBI-MFC (NL PPdE Production Center)" ;
            :nco_openmp_thread_number = 1 ;
            :Conventions = "CF-1.0" ;
            :references = "http://marine.copernicus.eu" ;
            :domain_name = "IBI36" ;
            :contact = "mailto: servicedesk.cmems@mercator-ocean.eu" ;
            :title = "Static files for product IBI_ANALYSISFORECAST_BGC_005_004" ;
            :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
            :licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-and-
licence/" ;
            :producer = "CMEMS - IBI Monitoring and Forecasting Centre" ;
            :institution = " Nologin (Spain) Puertos del Estado (PdE) - Mercator-Ocean (MO)" ;
            :area = "IBI" ;
            :product = "IBI_ANALYSISFORECAST_BGC_005_004" ;
            :dataset = "cmems_mod_ibi_bgc_anfc_0.027deg-3D_static " ;
            :product_user_manual = "http://marine.copernicus.eu/documents/PUM/CMEMS-IBI-
PUM-005-004.pdf" ;
            :quality_information_document =
"http://marine.copernicus.eu/documents/QUID/CMEMS-IBI-QUID-005-004.pdf" ;
    }

    netcdf IBI-MFC_005_004_coordinates {
    dimensions:
        latitude = 1081 ;
        longitude = 865 ;
        depth = 50 ;
    variables:
```

```
        float e1t(latitude, longitude) ;
                e1t:_FillValue = -32767.f ;
                e1t:long_name = "Cell dimension along X axis" ;
                e1t:units = "m" ;
        float e2t(latitude, longitude) ;
                e2t:_FillValue = -32767.f ;
                e2t:long_name = "Cell dimension along Y axis" ;
                e2t:units = "m" ;
        float depth(depth) ;
                depth:axis = "Z" ;
                depth:units = "m" ;
                depth:standard_name = "depth" ;
                depth:positive = "down" ;
        float e3t(depth, latitude, longitude) ;
                e3t:_FillValue = -32767.f ;
                e3t:long_name = "Cell dimension along Z axis" ;
                e3t:units = "m" ;
        float latitude(latitude) ;
                latitude:axis = "Y" ;
                latitude:standard_name = "latitude" ;
                latitude:units = "degrees_north" ;
                latitude:step = "0.02777863f" ;
        float longitude(longitude) ;
                longitude:axis = "X" ;
                longitude:standard_name = "longitude" ;
                longitude:units = "degrees_east" ;
                longitude:step = "0.02777863f" ;

    // global attributes:
                :Conventions = "CF-1.0" ;
                :domain_name = "IBI36" ;
                :contact = "mailto: servicedesk.cmems@mercator-ocean.eu" ;
                :references = "http://marine.copernicus.eu" ;
                :credit = "E.U. Copernicus Marine Service Information (CMEMS)" ;
                :licence = "http://marine.copernicus.eu/services-portfolio/service-commitments-and-
    licence/" ;
                :producer = "CMEMS - IBI Monitoring and Forecasting Centre" ;
                :area = "IBI" ;
                :title = "Static files for product IBI_ANALYSISFORECAST_BGC_005_004" ;
                :product = "IBI_ANALYSISFORECAST_BGC_005_004" ;
                :product_user_manual = "http://marine.copernicus.eu/documents/PUM/CMEMS-IBI-
    PUM-005-004.pdf" ;
                :dataset = "cmems_mod_ibi_bgc_anfc_0.027deg-3D_static " ;
                :institution = " Nologin (Spain) Puertos del Estado (PdE) - Mercator-Ocean (MO)" ;
                :quality_information_document =
    "http://marine.copernicus.eu/documents/QUID/CMEMS-IBI-QUID-005-004.pdf" ;
    }
```

The previous file structures shown as example correspond to the complete IBI files, which are downloaded through FTP. However, when data from the IBI forecast product are downloaded through the Subsetter interface, the file map and structure change slightly, and it is dependent on the parameter selection made by the user in the specific data request. downloaded with the original file format.